

No Centrally Symmetric Hexagon Attains the Planar L_2 Rogers–Shephard Bound

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Abstract

For the Firey sum at $p = 2$, we prove that every full-dimensional centrally symmetric hexagon $K \subset \mathbb{R}^2$ containing the origin satisfies

$$|K \oplus_2 (-K)| < \left(2 + \frac{\pi}{2}\right) |K|.$$

The center of symmetry is arbitrary, and the result includes placements for which the origin lies on an edge or at a vertex. The proof parametrizes the complete affine hexagon family, reduces all origin placements to three vertex types by a shadow-system convexity argument, and evaluates those three areas exactly. This is a fixed- p , six-vertex result only; the equality classification for all centrally symmetric planar bodies and all $p > 1$ remains unresolved.

1 Statement and context

For a compact convex set $A \subset \mathbb{R}^2$, write

$$h_A(u) = \max_{x \in A} x \cdot u$$

for its support function. When $0 \in A \cap B$, their Firey L_2 sum is defined by

$$h_{A \oplus_2 B}(u)^2 = h_A(u)^2 + h_B(u)^2.$$

Fradelizi, Manui, Meyer, and Ndiaye proved that a centrally symmetric planar convex body K containing the origin satisfies

$$|K \oplus_p (-K)| \leq \left(2 + \frac{2\Gamma(1 + 1/q)^2}{\Gamma(1 + 2/q)}\right) |K|, \quad \frac{1}{p} + \frac{1}{q} = 1,$$

and that a parallelogram having the origin as a vertex attains equality [1, Corollary 29]. Their Conjecture 5 asks whether these are the only equality cases. At $p = q = 2$, the constant is $2 + \pi/2$.

A centrally symmetric quadrilateral is a parallelogram, so centrally symmetric hexagons form the smallest non-parallelogram polygonal family. The result of this note excludes that entire family at $p = 2$.

Theorem 1. *Let $K \subset \mathbb{R}^2$ be a full-dimensional centrally symmetric hexagon. If $0 \in K$, then*

$$|K \oplus_2 (-K)| < \left(2 + \frac{\pi}{2}\right) |K|.$$

No assumption is made on whether 0 is an interior or boundary point of K .

Theorem 1 neither treats polygons with eight or more vertices or nonpolygonal bodies, nor treats $p \neq 2$. It therefore does not settle the general equality classification.

2 Affine normalization and the placement variable

Write $K = c + S$, where $S = -S$ and c is the center of symmetry of K . Since S is symmetric,

$$0 \in c + S \iff -c \in S \iff c \in S. \quad (1)$$

Every centrally symmetric hexagon is the zonotope generated by its three pairs of parallel opposite edges. Choose two extreme generator directions. After changing generator signs and applying a nonsingular linear map, the third direction is a positive linear combination of the first two. Thus the complete six-vertex family has the form

$$S = \frac{1}{2}([-e_1, e_1] + [-e_2, e_2] + [-v, v]), \quad v = (a, b), \quad a, b > 0. \quad (2)$$

Conversely, (2) has exactly six vertices whenever $a, b > 0$. Nonsingular linear maps commute with Firey addition and multiply both areas by the same positive factor in absolute value. Hence this normalization preserves the ratio under study.

Put

$$\begin{aligned} D = |S| &= 1 + a + b, \\ r_1 &= \frac{1}{2}(1 - a, -1 - b), \\ r_2 &= \frac{1}{2}(1 + a, b - 1), \\ r_3 &= \frac{1}{2}(1 + a, 1 + b). \end{aligned} \quad (3)$$

The six vertices, in cyclic order, are $-r_3, r_1, r_2, r_3, -r_1, -r_2$. In view of (1), the full compact set $c \in S$ parametrizes every admissible placement of the origin, including the boundary.

For $L_c = (c + S) \oplus_2 (-c - S)$, direct expansion gives

$$h_{L_c}(u) = \sqrt{2} \sqrt{h_S(u)^2 + (c \cdot u)^2}. \quad (4)$$

Lemma 2 (Placement convexity). *The map $c \mapsto |L_c|$ is convex on every affine line. Consequently, for $c \in S$,*

$$|L_c| \leq \max\{|L_{r_1}|, |L_{r_2}|, |L_{r_3}|\}. \quad (5)$$

Proof. Define

$$E_c = \text{conv}\{\alpha x + \beta c : x \in S, \alpha^2 + \beta^2 \leq 1\}.$$

The symmetry of S implies

$$\begin{aligned} h_{E_c}(u) &= \max_{\alpha^2 + \beta^2 \leq 1} (|\alpha| h_S(u) + \beta(c \cdot u)) \\ &= \sqrt{h_S(u)^2 + (c \cdot u)^2}. \end{aligned}$$

Thus (4) says that $L_c = \sqrt{2}E_c$.

Consider $c_t = c_0 + tw$. Every generating point of E_{c_t} is

$$(\alpha x + \beta c_0) + t\beta w,$$

so all generating points move parallel to w , and the projection of E_{c_t} onto $w^\perp$ is independent of t . In a fixed projected fibre, the upper endpoint is the supremum of affine functions of t ; the negative of the lower endpoint has the same property. The fibre length is therefore convex in t . Integration over the fixed projection shows that $|E_{c_t}|$, and hence $|L_{c_t}|$, is convex.

A convex function on the polygon S is bounded above by its largest value at a vertex. Formula (4) is even in c , so opposite vertices give the same value, leaving the three representatives in (5). $\square$

3 Exact evaluation at the three vertex types

Use projective normal coordinates $n(t) = (1, t)$ on the right half of the normal circle. From (2) and (3),

$$h_S(n(t)) = \begin{cases} r_1 \cdot n(t), & t \leq -a/b, \\ r_2 \cdot n(t), & -a/b \leq t \leq 0, \\ r_3 \cdot n(t), & t \geq 0. \end{cases} \quad (6)$$

On an interval with active vertex r , let $A = rr^\top + cc^\top$. The boundary point selected by $n(t)$ is

$$x(t) = \frac{\sqrt{2An(t)}}{\sqrt{n(t)^\top An(t)}}.$$

Because $\det A = \det(r, c)^2$, differentiation yields

$$\frac{1}{2} \int \det(x(t), x'(t)) dt = \det(A) \int \frac{dt}{n(t)^\top An(t)} = |\det(r, c)| \left| \Delta \arg((r \cdot n) + i(c \cdot n)) \right|. \quad (7)$$

The last expression is interpreted as zero when r and c are parallel; it is also the continuous rank-one limit of the preceding integral.

Applying (7) on the three intervals in (6) gives the following contributions from the smooth arcs on this half of the boundary:

c	$(-\infty, -a/b)$	$(-a/b, 0)$	$(0, \infty)$
r_1	0	$\frac{a+b}{2} \arctan a$	$\frac{1+b}{2} \arctan(1/a)$
r_2	$\frac{a+b}{2} \arctan b$	0	$\frac{1+a}{2} \arctan(1/b)$
r_3	$\frac{1+b}{2} \arctan(b/a)$	$\frac{1+a}{2} \arctan(a/b)$	0.

(8)

There are also boundary segments at the changes of active vertex. Suppose adjacent signed vertices r, r' satisfy $r \cdot n = r' \cdot n = \rho > 0$, and put $d = c \cdot n$. The two limiting boundary points show that the segment's oriented area contribution is

$$\frac{(\rho r + dc) \times (\rho r' + dc)}{\rho^2 + d^2}, \quad (9)$$

where $x \times y = \det(x, y)$. On the chosen half of the normal circle, the three transitions are

$$(r_1, r_2; (b, -a)), \quad (r_2, r_3; (1, 0)), \quad (r_3, -r_1; (0, 1)).$$

For each $c \in \{r_1, r_2, r_3\}$, substitution into (9) gives, respectively,

$$\frac{a+b}{2}, \quad \frac{1+a}{2}, \quad \frac{1+b}{2}.$$

Their sum is D . The body L_c is centrally symmetric, so doubling the smooth-arc contributions in (8) and the segment sum gives

$$|L_{r_1}| = 2D + (a+b) \arctan a + (1+b) \arctan(1/a), \quad (10)$$

$$|L_{r_2}| = 2D + (a+b) \arctan b + (1+a) \arctan(1/b), \quad (11)$$

$$|L_{r_3}| = 2D + (1+b) \arctan(b/a) + (1+a) \arctan(a/b). \quad (12)$$

4 Strict deficits and proof of the theorem

For $x > 0$, $\arctan x + \arctan(1/x) = \pi/2$. Subtracting (10)–(12) from $(2 + \pi/2)D$ gives the exact deficits

$$\delta_1 = \arctan a + a \arctan(1/a), \quad (13)$$

$$\delta_2 = \arctan b + b \arctan(1/b), \quad (14)$$

$$\delta_3 = b \arctan(a/b) + a \arctan(b/a). \quad (15)$$

Every term in (13)–(15) is positive because $a, b > 0$. Lemma 2 therefore yields, in the normalized coordinates,

$$\left(2 + \frac{\pi}{2}\right) |S| - |L_c| \geq \min\{\delta_1, \delta_2, \delta_3\} > 0 \quad (c \in S).$$

Undoing the nonsingular linear normalization multiplies this strict deficit by a positive factor. This proves Theorem 1.

The positive gap is not uniform over the full open shape family. For example, it can tend to zero when $a \downarrow 0$, as the hexagon degenerates to a parallelogram. Such a limit does not omit a genuine six-vertex member of the normalized family.

Computational and AI assistance

An accompanying SymPy script checks the determinant identities used in (8), the three transition contributions in (9), and the reductions (13)–(15). These symbolic checks do not establish the affine coverage, placement convexity, or positivity arguments, which are proved above. An AI system assisted with proof exploration, bounded source retrieval, exposition, and generation of the symbolic verification code; the author reviewed the mathematical argument and takes responsibility for the claims. The user-supplied unpublished item on AI code generation [2] was considered only in framing this assistance disclosure. It is not evidence for any mathematical assertion, prior-art assessment, or status statement in this note.

References

- [1] M. Fradelizi, A. Manui, M. Meyer, and C. S. Ndiaye, *L_p -Rogers–Shephard type inequalities for L_p -zonoids and symmetric bodies*, arXiv:2607.03582v1, 2026.
- [2] Zijian Zeng, *AI Chatbot for Code Generation*, user-supplied unpublished manuscript, 2025.